

Introduction to Many-Body Quantum Information

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Created: March 4, 2026

Updated: May 9, 2026

3 Bipartite entanglement (I)

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Main references for this section:

- Michael A. Nielsen, Isaac L. Chuang. *Quantum Computation and Quantum Information*.
- Martin B. Plenio, S. Virmani. Quant.Inf.Comput.7:1-51, (2007).

3.1 Quantum operations

In quantum information theory, the most general physical transformation of a quantum state is described by a *quantum operation*², i.e., a *completely positive trace-preserving (CPTP) map*

$$\mathcal{E} : \rho \longrightarrow \mathcal{E}(\rho); \quad (66)$$

- *Trace-preserving (TP)*. The trace-preserving condition ensures probability conservation:

$$\text{Tr}(\mathcal{E}(\rho)) = \text{Tr}(\rho) = 1. \quad (67)$$

- *Positivity*. A map $\mathcal{E}$ is *positive* if it maps any density matrix to another positive operator:

$$\rho \geq 0 \Rightarrow \mathcal{E}(\rho) \geq 0. \quad (68)$$

- *Complete positivity (CP)*. Physical consistency requires a stronger condition: CP. A map is CP if, for any auxiliary system R ,

$$(\mathcal{E} \otimes I_E)(\rho_{SE}) \geq 0 \quad (69)$$

for all joint states ρ_{SE} .

This condition guarantees that the map remains physical even when acting on part of an entangled system.

²Also known as *quantum map* or *quantum channel*.

An important example of positivity but not CP is the *partial transpose* map $\rho_{SE}^{T_E}$, where T_E denotes matrix transposition for environment E . As a result, T_E can map a valid density matrix to a non-positive operator.

For example, consider the Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \quad (70)$$

with density matrix

$$\begin{aligned} \rho &= |\Phi^+\rangle \langle \Phi^+| \\ &= \frac{1}{2} (|00\rangle \langle 00| + |00\rangle \langle 11| + |11\rangle \langle 00| + |11\rangle \langle 11|). \end{aligned} \quad (71)$$

By taking partial transpose to the second qubit, we obtain

$$\begin{aligned} \rho^{T_2} &= \frac{1}{2} (|00\rangle \langle 00| + |01\rangle \langle 10| + |10\rangle \langle 01| + |11\rangle \langle 11|) \\ &= \begin{bmatrix} 1/2 & 0 & 0 & 0 \\ 0 & 0 & 1/2 & 0 \\ 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1/2 \end{bmatrix} \end{aligned} \quad (72)$$

One can easily obtain its eigenvalues $\{1/2, 1/2, 1/2, -1/2\}$, which are not fully positive.

3.2 Kraus representation

Theorem 3.1. *Any CPTP map admits a Kraus representation:*

$$\mathcal{E}(\rho) = \sum_i K_i \rho K_i^\dagger, \quad (73)$$

where the Kraus operators satisfy the completeness condition

$$\sum_i K_i^\dagger K_i = I. \quad (74)$$

Physically, this describes a process where different outcomes i occur with probabilities

$$p_i = \text{Tr}(K_i \rho K_i^\dagger), \quad (75)$$

and the corresponding post-measurement states are

$$\rho_i = \frac{K_i \rho K_i^\dagger}{p_i}. \quad (76)$$

Note that the Kraus operators have some unitary freedom. Consider a unitary transformation U such that

$$L_\alpha = \sum_i u_{\alpha i} K_i \Leftrightarrow L_\alpha^\dagger = \sum_j u_{\alpha j}^* K_j^\dagger. \quad (77)$$

Then the quantum channel described by the new operators $\{L_\alpha\}$ is

$$\begin{aligned}
\mathcal{E}'(\rho) &= \sum_{\alpha} L_{\alpha} \rho L_{\alpha}^{\dagger} \\
&= \sum_{\alpha} \left(\sum_i u_{\alpha i} K_i \right) \rho \left(\sum_j u_{\alpha j}^* K_j^{\dagger} \right). \\
&= \sum_{i,j} \left(\sum_{\alpha} u_{\alpha i} u_{\alpha j}^* \right) K_i \rho K_j^{\dagger} = \sum_i K_i \rho K_i^{\dagger} = \mathcal{E}(\rho),
\end{aligned} \tag{78}$$

where we have used

$$UU^{\dagger} = I \Leftrightarrow \sum_{\alpha} u_{\alpha i} u_{\alpha j}^* = \delta_{ij}. \tag{79}$$

Remark 3.2. For a bipartite system AB , a *local quantum operation* acts only on one subsystem, e.g.

$$\mathcal{E}_A(\rho_{AB}) = \sum_i (K_i \otimes I) \rho_{AB} (K_i^{\dagger} \otimes I). \tag{80}$$

Similarly for subsystem B .

3.3 Local operations and classical communication

We begin by recalling the definitions of bipartite separable and entangled states:

Definition 3.3. A bipartite state ρ_{AB} is said to be *separable* if it can be written as

$$\rho_{AB} = \sum_k p_k \rho_A^{(k)} \otimes \rho_B^{(k)}, \quad p_k \geq 0, \quad \sum_k p_k = 1. \tag{81}$$

In particular, a pure separable state is called a *product state*. States that cannot be expressed in this form are called *entangled states*.

A bipartite separable state is said to have zero *entanglement*, while an entangled state is said to have nonzero entanglement.

This motivates us to find some measure to quantify the amount of entanglement in a given state.

Definition 3.4 (*Local operations and classical communication (LOCC)*). LOCC refers to protocols in which spatially separated parties are restricted to performing local quantum operations on their respective subsystems, while coordinating their actions through classical communication.

Proposition 3.5. Any separable state can be prepared using LOCC.

This observation provides the key physical motivation for defining a *measure of entanglement*:

- LOCC operations cannot create entanglement $\Rightarrow$ a valid entanglement measure $E(\rho)$ is required to be non-increasing under LOCC.

This property is called *monotonicity*, and we would say E is an *entanglement monotone*. Mathematically, for any deterministic LOCC channel $\mathcal{E}_{\text{LOCC}}$, one requires

$$E(\rho) \geq E(\mathcal{E}_{\text{LOCC}}(\rho)). \quad (82)$$

Second, for a general *probabilistic LOCC* protocol producing an ensemble $\{p_i, \rho_i\}$, the stronger *average monotonicity* condition should hold:

$$E(\rho) \geq \sum_i p_i E(\rho_i). \quad (83)$$

This ensures that even when conditioning on measurement outcomes (*postselection*), entanglement cannot be increased on average.

3.4 Schmidt decomposition and entanglement entropy

We begin with discussing bipartite *pure states*, for which entanglement admits a particularly simple and complete measure—the *entanglement entropy*.

Theorem 3.6 (Schmidt decomposition). *Let $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ be a bipartite pure state. Then there exist orthonormal bases $\{|i_A\rangle\}$ of $\mathcal{H}_A$ and $\{|i_B\rangle\}$ of $\mathcal{H}_B$ such that*

$$|\psi\rangle = \sum_{i=1}^r \lambda_i |i_A\rangle \otimes |i_B\rangle, \quad (84)$$

*where $\lambda_i \geq 0$ are called **Schmidt coefficients** with $\sum_i \lambda_i^2 = 1$, and r is called the **Schmidt rank**.*

Proof. Choose orthonormal bases $\{|j\rangle\}$ for $\mathcal{H}_A$ and $\{|k\rangle\}$ for $\mathcal{H}_B$, and expand

$$|\psi\rangle = \sum_{jk} a_{jk} |j\rangle \otimes |k\rangle. \quad (85)$$

By applying the *singular value decomposition*,

$$a = u d v = \sum_{ijk} u_{ji} d_{ii} v_{ik}, \quad (86)$$

where d is a diagonal matrix with elements $\{d_{ii}\}$, we have

$$|\psi\rangle = \sum_i \lambda_i |i_A\rangle \otimes |i_B\rangle, \quad (87)$$

with

$$|i_A\rangle := \sum_j u_{ji} |j\rangle, \quad |i_B\rangle := \sum_k v_{ik} |k\rangle, \quad \lambda_i := d_{ii}. \quad (88)$$

□

The Schmidt decomposition expresses any pure state as a sum of *paired* orthonormal states, with non-negative coefficients $\{\lambda_i\}$ uniquely determined (up to degeneracies).

Corollary 3.7. *The state is a product state if and only if $r = 1$.*

Using the Schmidt decomposition, one finds, the reduced density matrices satisfy

$$\rho_A = \sum_i \lambda_i^2 |i_A\rangle \langle i_A|, \quad \rho_B = \sum_i \lambda_i^2 |i_B\rangle \langle i_B|. \quad (89)$$

Thus, ρ_A and ρ_B share the same nonzero eigenvalues $\{\lambda_i^2\}$.

Note that $\lambda_i \in [0, 1]$ and $\sum_i \lambda_i^2 = 1$, so that $\{\lambda_i^2\}$ can be regarded as an effective probability distribution. When this distribution becomes a delta function, i.e., $\lambda_j = 1$ and $\lambda_{i \neq j} = 0$, the state is unentangled (separable). Otherwise, the state is entangled.

Definition 3.8. Motivated by the classical Shannon entropy, we define the *entanglement entropy*—also known as the *von Neumann entropy*—as

$$S(\rho_A) = -\text{Tr}(\rho_A \log \rho_A), \quad (90)$$

which quantifies the entanglement of a bipartite pure state.

In terms of the Schmidt coefficients, this becomes

$$S(\rho_A) = -\sum_i \lambda_i^2 \log \lambda_i^2. \quad (91)$$

Moreover, it satisfies:

- $S = 0$ if and only if the state is separable ($r = 1$);
- S is maximal when $\lambda_i^2 = 1/d$ for all i (maximally entangled state).

For a bipartite pure state $|\psi\rangle_{AB}$, we claim that the entanglement entropy S is an *entanglement measure* satisfying the following properties:

- *Faithfulness:*

$$S(\rho_A) = 0 \quad \text{if and only if} \quad |\psi\rangle_{AB} \text{ is separable.}$$

- *Symmetry*:

$$S(\rho_A) = S(\rho_B),$$

reflecting that entanglement is a property shared by both subsystems.

- *Invariance under local unitaries*: For any local unitary $U_A \otimes U_B$,

$$S(\rho_A) = S((U_A \otimes U_B) |\psi\rangle \langle\psi| (U_A^\dagger \otimes U_B^\dagger)).$$

- *Additivity*: For the case of $\rho_{AB} = \rho_A \otimes \rho_B$,

$$S(\rho_{AB}) = S(\rho_A) + S(\rho_B). \quad (92)$$

- *Continuity*: Small changes in the state lead to small changes in the entropy.

- *Monotonicity under LOCC*: Entanglement entropy does not increase under LOCC operations,

$$S(\mathcal{E}_{\text{LOCC}}(\rho)) \leq S(\rho). \quad (93)$$

- *Concavity*: As a function of the density matrix ³,

$$S\left(\sum_k p_k \rho_k\right) \geq \sum_k p_k S(\rho_k). \quad (94)$$

3.5 Rényi entropies

While the von Neumann entropy provides a complete characterization of entanglement for bipartite pure states, it is often useful to consider a broader family of entropy measures. A particularly important generalization is given by the *Rényi entropies*, which interpolate between different notions of quantum uncertainty.

Definition 3.9. For a density matrix ρ , the *Rényi entropy* of order $\alpha > 0, \alpha \neq 1$, is defined as

$$S_\alpha(\rho) := \frac{1}{1-\alpha} \log \text{Tr}(\rho^\alpha). \quad (95)$$

When $\alpha \rightarrow 1$, it recovers the von Neumann entropy,

$$\lim_{\alpha \rightarrow 1} S_\alpha(\rho) = -\text{Tr}(\rho \log \rho). \quad (96)$$

For a bipartite pure state $|\psi\rangle_{AB}$, the Rényi entropies of the reduced density matrix ρ_A also serve as entanglement measures. In terms of Schmidt coefficients $\{\lambda_i\}$,

$$S_\alpha(\rho_A) = \frac{1}{1-\alpha} \log \sum_i \lambda_i^{2\alpha}. \quad (97)$$

It satisfies

³This reflects that entropy is increased by classical mixing. The excess contribution does not originate from additional quantum correlations, but from the classical uncertainty associated with the probabilistic preparation.

- *Faithfulness*: $S_\alpha = 0$ if and only if the state is separable.
- *Symmetry*: $S_\alpha(\rho_A) = S_\alpha(\rho_B)$ for pure states.
- *Invariance under local unitaries*: also for the von Neumann entropy.
- *Entanglement monotones for bipartite pure state*.

Furthermore, we remark that the second Rényi entropy admits a particularly useful representation in terms of the *SWAP operator*.

Consider two copies of the system, with Hilbert space $\mathcal{H} \otimes \mathcal{H}$. Define the SWAP operator $\mathbb{S}$ acting as

$$\mathbb{S} |i\rangle |j\rangle = |j\rangle |i\rangle. \quad (98)$$

Then, for a density matrix ρ , one has

$$\text{Tr}(\rho^2) = \text{Tr} [(\rho \otimes \rho) \mathbb{S}]. \quad (99)$$

Therefore,

$$S_2(\rho) = -\log \text{Tr} [(\rho \otimes \rho) \mathbb{S}]. \quad (100)$$

More generally, for a bipartite system, the subsystem Rényi-2 entropy can be written as

$$\text{Tr}(\rho_A^2) = \text{Tr} [(\rho_{AB} \otimes \rho_{AB}) \mathbb{S}_A], \quad (101)$$

where $\mathbb{S}_A$ swaps only the degrees of freedom in subsystem A .

This representation provides a direct connection between Rényi entropies and measurable observables in replicated systems.